

RESEARCH

Open Access



Repetitive negative thinking is associated with cognitive function decline in older adults: a cross-sectional study

Niansi Ye^{1,2}, Ling Peng³, Bei Deng^{1,4,5}, Hui Hu^{1,4,5*}, Yuncui Wang^{1,4,5}, Taoyun Zheng^{1,4,5}, Yating Ai^{1,4,5}, Xueting Liu^{1,4,5}, Shi Zhou^{1,4,5} and Yucan Li^{1,4,5}

Abstract

Background Psychological problems such as depression and anxiety increase the risk of cognitive impairment in older adults. But mechanisms on the effect of psychological disorder on cognitive function is inconclusive. Repetitive negative thinking (RNT) is a core symptom of a number of common psychological disorders and may be a modifiable process shared by many psychological risk factors that contribute to the development of cognitive impairment. RNT may increase the risk of cognitive impairment. However, there are fewer studies related to RNT and cognitive function, and there is a lack of epidemiological studies to explore the relationship between RNT and cognitive function.

Methods A cross-sectional study of 424 older adults aged 60 years or over was performed from May to November 2023 in hospital. To investigate the RNT level by using the Perseverative Thinking Questionnaire (PTQ), and investigate the cognitive function level by using the Montreal Cognitive Assessment Scale (MoCA). Multivariable linear regression and subgroup analyses were used to explore the relationship between RNT and cognitive function.

Results We categorized the total RNT scores into quartiles. The multivariable linear regression analysis showed that after adjusting for all covariates, the participants in the Q3 and Q4 groups exhibited lower cognition scores (Q3: $\beta = -0.180$, 95%CI $-2.849 \sim -0.860$; Q4: $\beta = -0.164$, 95% $-2.611 \sim -0.666$) compared to the Q1 group. The results of the subgroup analyses showed that individuals aged 60 ~ 79 years, junior high school and above are more prone to suffer from cognitive impairment with a high RNT score.

Conclusion The study reveals a negative association between RNT and cognitive function in community-dwelling older adults. However, multi-center and a longer time span cohort studies on the relationship between RNT and cognitive function should be carried out to further explore the mechanisms involved.

Keywords Repetitive negative thinking, Cognitive function, Older adults, Association

*Correspondence:

Hui Hu
huihu90@hbu.cn

¹School of Nursing, Hubei University of Chinese Medicine, Wuhan, China

²Nursing Department, Wuhan No.1 Hospital, Wuhan, China

³Department of Neurosurgery, Tongji Hospital, Tongji Medical College, Hua Zhong University of Science and Technology, Wuhan, China

⁴Engineering Research Center of TCM Protection Technology and New Product Development for the Elderly Brain Health, Ministry of Education, Wuhan, China

⁵Hubei Shizhen Laboratory, Wuhan, China



© The Author(s) 2025, corrected publication 2025. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

In the context of a grim situation of population aging and a prominent trend of advanced aging in China, cognitive impairment has become one of the major diseases that seriously endanger the health of the older adults and affect the sustainable development of the society, which has a negative impact on the physical and psychological health as well as the quality of life of the older adults. The onset of cognitive impairment begins with the age-related declines in cognitive function, progresses to mild cognitive impairment (MCI), and ends with dementia [1]. The patients unable to live independently in the later stages of the disease. Around 55 million people worldwide currently suffer from dementia, with the figure expected to reach 139 million by 2050 [2]. The prevalence of dementia in China aged 60 years and over is 6.0%, and the prevalence of MCI is 15.5% [3]. The prevalence of cognitive disorders is increasing year by year, placing a heavy burden on patients, families, and society. It is estimated that the total annual cost of dementia disease in China will reach \$1.89 trillion in 2050 [4]. However, there is no drug that can stop or reverse the progression of dementia. Cognitive decline can be effectively prevented or delayed by controlling risk factors at an early stage, so it is important to identify, prevent, and treat risk factors associated with cognitive impairment [5].

With the rapid development of society, the transformation of family structures, and the decline of physiological functions brought about by aging, the psychological health problems of the older adults have become increasingly prominent [6, 7]. Anxiety and depression are the most common psychological health problems among the older adults, and the probability of suffering from depressive symptoms is currently 22.6% and that of suffering from anxiety symptoms is 22.11% in China [8, 9]. Various physical illnesses caused by psychological problems in older adults threaten their physical and psychological health and the quality of existence. Psychological problems such as depression and anxiety have been found to increase the risk of cognitive impairment in older adults [10, 11]. But mechanisms on the effect of psychological disorder on cognitive function is inconclusive [12, 13].

RNT includes rumination and worry. Rumination refers to a maladaptive response style, which is characterised by repeated and unconscious passive thinking about the causes, consequences and effects of negative life events, and a persistent preoccupation with negative experiences rather than taking positive practical action [14]. Worry describes repetitive thoughts about potential threats, uncertain events, and risky events in the future [15]. The main difference between rumination and worry is in time and content [16]. It has been found that heightened levels of rumination and/or worry are present in the most Axis I disorder, including 13 categories

of psychological disorders such as depression, anxiety disorders, sleep disorders, and post-traumatic stress disorder [17–19]. Based on the widespread presence of rumination and worry across disorders, it has been suggested that RNT is a transdiagnostic process that shows the same characteristics across disorders, whereby only the content is disorder-specific [20]. RNT is the repetitive thinking about one or more negative issues that are difficult to control [21]. Research has shown that RNT is a core symptom of depression, anxiety, and many other common psychological illnesses. The higher levels of RNT lead to increased susceptibility to a wide range of mood disorders [22]. RNT as a common process in Axis I disorder may be a common pathway for psychological disorder leading to an increased risk of dementia. Therefore, we propose that RNT may be a modifiable process for many of the psychological risk factors that contribute to cognitive decline and it increase the risk of cognitive decline.

RNT has been found to correlate with dementia biomarkers, global cognition, and subjective cognitive decline in older adults [23, 24]. Although these studies provide evidence of a relationship between RNT and poorer objective and subjective cognition. Since differences in the study populations and assessment methods of different studies may have an impact on the results, we aimed to explore the association between RNT and cognitive function in community-dwelling older adults in China to provide evidence for the prevention of cognitive decline.

Methods

Population

In this study, a cross-sectional study method was used to select participants from community in Wuhan. The sample size was calculated using the formula of $[Z^2P(1-P)]/d^2$, at level of significance at 0.05 and CI of 95% [25]. The prevalence of cognitive disorder in the Wuhan was taken at a level of 0.878 with a relative precision of 0.05 [26]. A sample size of 190 participant was estimated to assess its correlation with RNT with a potential dropout rate of 15%. The questionnaire survey was conducted from May 2023 to November 2023 among 424 participants in the community of Wuhan. The inclusion criteria for participants included: (1) age over 60 years; (2) a local residence time ≥ 6 months; (3) ability to communicate normal and complete the questionnaire; and (4) signed informed consent. Considering the impact of selected diseases on the cognitive function, the exclusion criteria included: (1) presence dementia such as Alzheimer's disease, vascular dementia, and other neurological diseases that can cause brain dysfunction, which diagnosed by a medical institution; (2) presence of severe heart, liver, and kidney diseases and malignant tumours; (3) alcohol, drug

abuse or dependence within the previous 2 years; (4) have psychological disease diagnosed by a medical institution. Prior to both the interviews and examinations, all participants provided informed consent. The study was approved by the Ethics Committee of Hubei University of Chinese Medicine (Approved No. of ethic committee: 2019-IEC-003).

Repetitive negative thinking assessment

RNT was assessed using the perseverative thinking questionnaire (PTQ). The scale consists of 15 items covering three domains: core characteristics of RNT, unproductiveness, and psychological capacity captured. Each item is rated on a 5-point Likert scale from 0 “never” to 4 “almost always”, with a total score ranging from 0 to 40. The higher score of PTQ represents higher levels of RNT. The Cronbach's α of PTQ is 0.95 [20]. The questionnaire is currently available in Chinese, German, English, Polish and French. Good reliability and validity when applied to the older adults, young people, children and women [27, 28].

Cognitive function assessment

Montreal Cognitive Assessment (MoCA) Test is a widely used screening assessment tool for cognitive function of older adults. Studies have shown that the MoCA test has high sensitivity (80-100%) and specificity (50-76%) in identifying MCI, and it is more accurate than the Mini-Mental State Examination Scale in distinguishing between normal and MCI (Grade A recommendation) [29]. The MoCA test measures a wider range of cognitive domains, including visuospatial abilities, executive functions, attention, memory, concentration, language, verbal abstraction, and orientation. There are a total of 11 test entries with a total score of 30, with higher scores indicating better cognitive function. One additional point was given to patients having <12 years of education for the MoCA scale. Cognitive function was assessed with MoCA (Beijing version). The Cronbach's α of MoCA is 0.818, which has a good measurement characteristic [30].

Covariates

In our study, covariates were used to mitigate potential confounding influences on the relationship between RNT and cognitive function, grounded on insights from prior research literature. These covariates included gender, age, occupation, marital status, living arrangement, education level, monthly income, and number of chronic disease, family history of Alzheimer's disease, and number of hobbies.

Statistical analysis

Quantitative data are presented as mean $\pm$ standard deviation, while qualitative data are expressed as numbers

(percentages). Data were tested for independence, normality, and homogeneity of the variances before statistical analyses. Independent samples t-test was used to compare the measurement data between the two groups. Multi-group comparison was determined by the one-way ANOVA or Welch's test as appropriate. If $p < 0.05$, the data of the two groups were considered to have statistical differences. Associations between normally distributed variables were analyzed using Pearson correlation. To examine the association between RNT and cognitive function, a linear regression model was conducted. In order to enrich the findings and provide clearer clinical implications, total RNT score was categorized based on quartiles (Q1: < 25th percentile, Q2: 25 to 50th percentile, Q3: 50 to 75th percentile, Q4: $\geq$ 75th percentile) with Q1 as the reference category. Furthermore, subgroup analyses were conducted based on factors such as age and educational level to investigate whether these factors influenced the relationship between RNT and cognitive function. A P -value < 0.05 was considered statistically significant. SPSS 25.0 was used for statistical analysis in this study. This was an exploratory analysis; thus, adjustment for multiple comparisons was not made.

Results

Participants characteristics

Table 1 presents participant characteristics. This analysis included 424 participants from Wuhan in Hubei Province. Of these participants, 161 (37.97%) were male and 263 (62.03%) were female, and the weighted mean age was 68.93 ± 0.26 years. Different age, occupation, marital status, living arrangement, education level, monthly income, and number of hobbies were significantly different across the cognitive function.

Cognitive function scores for comparison among the RNT quartiles

Table 2 presents the relationship between RNT and cognitive function. The results of the Pearson correlation analyses showed that RNT is associated with global cognition and cognitive domains except language skills.

Table 3 presents the comparison of the cognitive function in RNT quartiles. The interquartile ranges of RNT scores were 0 to 5, 6 to 12, 13 to 21.75, and 21.75 to 47, respectively. After stratifying the RNT, MoCA scores and cognitive domains score revealed differences in RNT quartiles. Participants in the Q3 and Q4 groups exhibited lower MoCA scores, visuospatial function score, naming score, abstracting score, memory score ($P < 0.05$).

Association between the RNT and cognitive function: results of regression analysis

Table 4 presents the findings of multivariable linear regression analysis on the association between RNT

Table 1 Comparison of participant characteristics

Variables	Overall, n = 424 (100.0%)	P Value
Age (year)		<0.001
60~69	167(39.39)	
70~79	177(41.75)	
80~90	80(18.87)	
Gender		0.482
Male	161(37.97)	
Female	263(62.03)	
Occupation		<0.001
Intellectual work	249(58.73)	
Manual work	148(34.91)	
Unemployed	27(6.37)	
Marital status		<0.001
Married	331(78.07)	
Divorced, widowed, and unmarried	93(21.93)	
Living arrangement		0.002
Living with spouse and children	101(23.82)	
Living with spouse	195(45.99)	
Living with children	75(17.69)	
Living alone	53(12.50)	
Education level		<0.001
Illiteracy	22(5.19)	
Primary school	53(12.50)	
Junior high school	115(27.12)	
High school and above	234(55.19)	
Monthly income (RMB)		<0.001
>8000	16(3.77)	
6000~	41(9.67)	
4000~	125(29.48)	
<4000	242(57.08)	
Number of chronic diseases		0.408
0	76(17.92)	
1	151(35.61)	
2	114(26.89)	
3	57(13.44)	
≥ 4	26(6.13)	
Family history of Alzheimer's disease		0.502
No	384(90.57)	
Yes	40(9.43)	
Number of hobbies		<0.001
≥ 4	8(1.89)	
3	162(38.21)	
2	151(35.61)	
1	72(16.98)	
0	31(7.31)	

Note. Percentages have been rounded and may not sum 100

All data were normally distributed. Age, occupation, living arrangement, education level, monthly income (RMB), number of hobbies didn't satisfy the homogeneity of variance, and were analyzed by Welch's test. The number of chronic diseases satisfied the homogeneity of variance, and was analyzed One-Way ANOVA. Gender, Marital status, and Family history of Alzheimer's disease met independence, normality, and variance homogeneity, which were determined by independent-Samples T tests.

and cognitive function. All regressions passed independence, normality test, and homogeneity of variances. Our research indicated that RNT was negatively associated with cognitive scores. The association remained statistically significant across all multivariate linear regression models, even after controlling for various covariates such as age, occupation, marital status, living arrangement, education level, monthly income, and number of hobbies. Age, education level, and RNT retained their statistical significance when entered into the final regression model. In Model 2, the participants in the Q3 and Q4 groups exhibited lower cognition scores (Q3: $\beta = -0.180$, 95%CI $-2.849 \sim -0.860$; Q4: $\beta = -0.164$, 95% $-2.611 \sim -0.666$) compared to the Q1 group.

Subgroup analysis

The final variables included in Model 2 included the variables age and education level in addition to RNT. Therefore, we want to further explore whether there is a correlation between RNT and cognition within different subgroups, including age (60~69 vs. 70~79 vs. ≥ 80~90) and education level (Illiteracy vs. Primary school vs. Junior high school vs. High school and above). The covariates included were those that were meaningful in the univariate analysis (age, occupation, marital status, living arrangement, education level, monthly income, and number of hobbies). The outcomes are displayed in Table 5. After adjusting for potential confounders, it was observed that RNT was negatively associated with cognitive function in the 60~79, middle school, and high school/technical school/secondary school groups. Within these subgroups, higher RNT scores were related to lower cognitive function scores. In contrast, RNT was not associated with cognitive function in the 80~90, primary school, and illiteracy.

Discussion

The present study suggested that the risk of cognitive impairment increased with higher RNT scores among older adults, and the robustness of the finding was confirmed through adjustment for various potential confounding variables. Additionally, individuals aged 60~79 years, junior high school and above were more prone to suffer from cognitive impairment with a high RNT score. However, the correlation between RNT and cognitive function was not significant in older adults aged 80 to 90 years, or those in elementary school and below.

To date, there have been limited endeavors to explore the correlation between RNT and cognitive function in older adults. Marchant et al. conducted a cohort study in 2016 to find that RNT was associated with decline in cognition, including global cognition, immediate and delayed memory [23]. In addition, the study found that increased level of RNT was associated with cognitive decline and

Table 2 The correlations between RNT and cognitive function

	MoCA	Visuospatial function	Naming	Attention	Language skills	Abstracting	Memory	orientation
RNT	−0.261**	−0.178**	−0.255**	−0.142**	−0.034	−0.178**	−0.149**	−0.122*

All data passed normality test

Note. ** $p < 0.01$; * $p < 0.05$ **Table 3** Comparison of the cognitive function in RNT quartiles

Variables	Q 1 (0 ~ 5) (N = 137)	Q 2 (6 ~ 12) (N = 84)	Q 3 (13 ~ 21.75) (N = 97)	Q4 (21.76 ~ 47) (N = 106)	P Value
MoCA	23.77 ± 4.482	23.23 ± 3.622	21.59 ± 4.138	21.46 ± 4.378	<0.001
Visuospatial function	3.69 ± 1.413	3.77 ± 1.123	3.28 ± 1.344	3.20 ± 1.348	0.003
Naming	2.66 ± 0.635	2.52 ± 0.702	2.25 ± 0.890	2.25 ± 0.916	<0.001
Attention	5.71 ± 1.677	5.71 ± 1.564	5.41 ± 1.712	5.24 ± 1.613	0.087
Language skills	1.73 ± 0.781	1.62 ± 0.727	1.67 ± 0.875	1.71 ± 0.828	0.778
Abstracting	1.36 ± 0.706	1.33 ± 0.646	1.08 ± 0.702	1.09 ± 0.775	0.002
Memory	2.75 ± 1.599	2.55 ± 1.555	2.21 ± 1.414	2.25 ± 1.542	0.020
Orientation	5.76 ± 0.576	5.61 ± 0.581	5.57 ± 0.762	5.58 ± 0.742	0.093

Assumptions of normality and independence were met for all significant data. MoCA, Visuospatial function, Naming, Language skills, Abstracting, Memory, and Orientation satisfied the homogeneity of variance and were analyzed by one-way ANOVA. Attention didn't satisfy the homogeneity of variance and was analyzed by Welch's test

Table 4 β values of RNT in linear regression models between RNT and cognitive scores after adjusting for different covariates

Model	B	β	t	95%CI	P Value
Model 1					
Total RNT	−0.087	−0.207	−4.839	−0.123 ~ −0.052	<0.001
Categories					
Q1	1(reference)	1(reference)	1(reference)	1(reference)	
Q2	−0.535	−0.049	−1.021	−1.564 ~ 0.495	0.308
Q3	−2.006	−0.195	−3.994	−2.994 ~ −1.019	<0.001
Q4	−1.823	−0.182	−3.687	−2.794 ~ −0.851	<0.001
Model 2					
Total RNT	−0.082	−0.194	−4.523	−0.117 ~ −0.046	<0.001
Categories					
Q1	1(reference)	1(reference)	1(reference)	1(reference)	
Q2	−0.315	−0.029	−0.597	−1.351 ~ 0.721	0.551
Q3	−1.855	−0.180	−3.664	−2.849 ~ −0.860	<0.001
Q4	−1.638	−0.164	−3.312	−2.611 ~ −0.666	<0.001

Model 1: age and education level were adjusted. Model 2: age, occupation, marital status, living arrangement, education level, monthly income, and number of hobbies were adjusted

Table 5 Subgroup analyses for RNT and cognitive function

Variable	B	β	t	P Value
Age				
60 ~ 69	−0.073	−0.196	−2.883	0.004
70 ~ 79	−0.101	−0.241	−3.482	<0.001
80 ~ 90	−0.046	−0.097	−0.979	0.331
Education Level				
Illiteracy	0.000	−0.001	−0.005	0.996
Primary school	−0.096	−0.274	−1.784	0.081
Junior high school	−0.079	−0.203	−2.180	0.031
High school and above	−0.078	−0.202	−3.182	0.001

neuroimaging biomarkers of Alzheimer's disease (i.e., amyloid, tau). Another cross-sectional study found that increased level of RNT was associated with worse subjective cognition and increased memory complaints.

Consistent with previous studies, our data demonstrates that higher level of RNT is related to worse cognitive function. In addition to this, this study found RNT was associated with cognitive domains except language skills. When participants were stratified by age and education level, a notable negative correlation was observed between RNT and cognitive function among older adults aged 60~69 years or junior high school and above. There are reasons why RNT does not correlate with cognitive function in older adults who are 80 to 90 years of age or have elementary school or below as follows. Increased brain aging in this age group may have altered the relationship between cognition and mood, or it may have weakened the association. The limited ability of older adults with low education level to perceive and express

RNT resulted in a non-significant correlation between RNT and cognitive function.

The underlying mechanisms linking psychological disorder to cognitive function remain vague. One study found that increased level of RNT was closely related to gray and white matter structures in the brain, particularly in the dorso-lateral prefrontal cortex, anterior cingulate cortex, the arcuate fasciculus, and superior longitudinal fasciculus [31]. These regions are related to cognitive control, emotion processing and regulation [32]. Increased level of RNT may lead to changes in the brain's structural functions related to cognitive control, leading to further cognitive decline. Cognitive debt theory suggests that psychology disorder can lead to damage to the hippocampus by increasing glucocorticoid levels and inducing inflammation and vascular disease in the brain, which impairs cognitive function [33]. RNT as a common trait of many types of psychology disorder, can be initiated and maintained without external triggers or awareness and narrows the scope of attention to repeatedly activated negative thoughts, thus provoking the individual to repeatedly experience physical and psychological distress, leading to the onset of psychology disorders, which in turn may increase the risk of cognitive impairment. As a person adopts the habits of negative thinking for a long-term, it constantly depletes the brain's limited resources, leading to a decline in the brain's ability to attention, executive functions, and memory [34, 35]. Older adulthood is a special stage with more pressure and stressful events. Along with the aging process, older adults will face physiological changes such as reduced self-care, frailty and the development of physical illnesses [36–38]. At the same time, they will experience negative stressful events such as a reduction in financial income, a decline in social status, and the death of friends and partners [39, 40]. These make older adults vulnerable to RNT, which further can have a range of negative effects on them.

Age is the biggest and uncontrollable risk factor for cognitive decline [41]. Literature has indicated that MCI incidence in China was 11.9% for older adults ages 60 to 69, 19.3% for 70 to 79, 24.4% for 80 to 89, 33.1% for 90 and above [3]. People over 80 are the fastest growing demographic around the world and they are at higher risk of developing cognitive impairment [42]. With the aging process, the physiological of the older adults gradually decline with the structure and function of the brain tissue gradual decline and the function of neural cell loss [43]. In addition, the continued accumulation of health risk factors increases the risk of chronic diseases such as hypertension, diabetes and coronary heart disease [44]. This disease led to amyloid plaque deposition through several mechanisms, such as increased oxidative stress, promoting inflammatory reaction, caused metabolic

disorders. These mechanisms increase the risk of cognitive decline.

Education level is a more consistent influence on cognitive function in most studies. Older adults with lower levels of education generally have limited nutritional conditions in early childhood or limited educational resources, which may have an impact on cognitive function [45]. They are more likely to be engaged in manual occupation and lack of exercise for brain, which leads to premature degeneration of neurons in the brain, thus reducing cognitive function [46]. In addition, older adults with lower levels of education may lack such knowledge, further increasing the risk of cognitive impairment [47].

This study offered multiple strengths. Firstly, in examining the association between RNT and cognitive function, the study eliminated as many bias-inducing factors as possible to ensure more reliable results through previous research and by conducting in-depth analyses that took into account a variety of possible potential confounders. Secondly, the study investigated the relationship between RNT and cognitive function through regression and subgroup analysis suggesting a negative association between RNT and cognitive function in community-dwelling older adults. In the future, the assessment of mental health can be incorporated into the health screening of older adults to comprehensively evaluate their health status. Health professionals and carers can enhance the assessment of RNT in older adults and identify problems promptly. By developing interventions to avoid further exacerbation of psychological problems in the elderly and increased risk of other diseases such as cognitive impairment.

However, there were limitations in the present study. First, a definitive causal relationship between RNT and cognitive function could not determine in this study since this study was a cross-sectional design. Secondly, since convenience sampling method was used in this study and all the participants in our study were selected only from Wuchan District and Hongshan District in Wuhan city, which suffered short time span, small sample size, and bad representativeness. In the future, multi-center and a longer time span cohort studies on the relationship between RNT and cognitive function should be carried out to further explore the mechanisms involved. Nonetheless, these findings have implications that are crucial to interventions that promote cognitive function in older adults.

Conclusion

In conclusion, this is the first study to investigate the relationship between RNT and cognitive function in Chinese older adults. After adjusting for a range of confounders, RNT is associated with cognitive function decline in older adults. The assessment of RNT levels in older

adults can be enhanced, and psychological interventions and other measures can be taken to reduce RNT levels and further prevent cognitive decline.

Abbreviations

RNT	Repetitive negative thinking
PTQ	Perseverative Thinking Questionnaire
MCI	Mild cognitive impairment
MoCA	Montreal Cognitive Assessment

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12888-025-06815-2>.

Supplementary Material 1

Acknowledgements

The authors thank all the study participants.

Author contributions

NSY involved in conception of study, acquisition of data, data entry, interpretation of results and drafting manuscript. LP and BD involved in acquisition of data and finalization of manuscript. HH involved in conception of study, acquisition of data, interpretation of results and finalization of manuscript. YCW, TYZ, and YTA involved in finalization of manuscript. XTL, SZ, and YCL involved in acquisition of data and data entry.

Funding

This study was funded by a National Natural Science Foundation of China in 2019(81973921). The authors appreciate the staff and residents of the institutions who participated in the study.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

The study was reviewed and approved by the Medical Ethics Committee of Hubei University of Chinese Medicine (Approved No. of ethic committee: [2019]IEC(003)). All methods were carried out in accordance with the relevant guidelines and regulations, following the principles of the Declaration of Helsinki. Informed consent was obtained, with subjects advised that participation was voluntary with information kept confidential.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Clinical trial number

Not applicable.

Received: 29 September 2024 / Accepted: 2 April 2025

Published online: 02 June 2025

References

- Canadian Task Force on Preventive Health Care, Pottie K, Rahal R, Jaramillo A, Birtwhistle R, Thombs BD, et al. Recommendations on screening for cognitive impairment in older adults. *CMAJ*. 2016;188(1):37–46. <https://doi.org/10.1503/cmaj.141165>.
- GBD 2019 Dementia Forecasting Collaborators. Estimation of the global prevalence of dementia in 2019 and forecasted prevalence in 2050: an analysis for the global burden of disease study 2019. *Lancet Public Health*. 2022;7(2):e105–25. [https://doi.org/10.1016/S2468-2667\(21\)00249-8](https://doi.org/10.1016/S2468-2667(21)00249-8).
- Jia L, Du Y, Chu L, Zhang Z, Li F, Lyu D, et al. Prevalence, risk factors, and management of dementia and mild cognitive impairment in adults aged 60 years or older in China: a cross-sectional study. *Lancet Public Health*. 2020;5(12):e661–71. [https://doi.org/10.1016/S2468-2667\(20\)30185-7](https://doi.org/10.1016/S2468-2667(20)30185-7).
- Jia J, Wei C, Chen S, Li F, Tang Y, Qin W, et al. The cost of Alzheimer's disease in China and re-estimation of costs worldwide. *Alzheimer's Dement J Alzheimer's Assoc*. 2018;14(4):483–91. <https://doi.org/10.1016/j.jalz.2017.12.006>.
- Livingston G, Sommerlad A, Orgeta V, Costafreda SG, Huntley J, Ames D, et al. Dementia prevention, intervention, and care. *Lancet*. 2017;390(10113):2673–734. [https://doi.org/10.1016/S0140-6736\(17\)31363-6](https://doi.org/10.1016/S0140-6736(17)31363-6).
- Zhou D, Zhan Q, Li L. The impact of self-employment on mental health of the younger elderly in China. *BMC Geriatr*. 2023;23(1):280. <https://doi.org/10.1186/s12877-023-03948-5>.
- Sivakumar PT, Mukku SSR, Antony S, Harbisetar V, Kumar CN, Math SB. Implications of mental healthcare act 2017 for geriatric mental health care delivery: A critical appraisal. *Indian J Psychiatry*. 2019;61(Suppl 4):S763–7. http://doi.org/10.4103/psychiatry.IndianJPsychiatry_100_19.
- Wang H, Hou Y, Zhang L, Yang M, Deng R, Yao J. Chinese elderly migrants' loneliness, anxiety and depressive symptoms: the mediation effect of perceived stress and resilience. *Front Public Health*. 2022;10:998532. <https://doi.org/10.3389/fpubh.2022.998532>.
- Li CWY, Yao YT, Hu YD. Mental health status of the elderly in China and intervention suggestions. *China Med Herald*. 2021;18(15):192–6.
- Boyle LL, Porsteinsson AP, Cui X, King DA, Lyness JM. Depression predicts cognitive disorders in older primary care patients. *J Clin Psychiatry*. 2010;71(1):74–9. <https://doi.org/10.4088/JCP08m04724gry>.
- Gulpers BJA, Verhey FRJ, Eussen SJPM, Schram MT, de Galan BE, van Bortel MPJ, et al. Anxiety and cognitive functioning in the Maastricht study: A cross-sectional population study. *J Affect Disord*. 2022;319:570–9. <https://doi.org/10.1016/j.jad.2022.09.072>.
- Botto R, Callai N, Cermelli A, Causarano L, Rainero I. Anxiety and depression in Alzheimer's disease: a systematic review of pathogenetic mechanisms and relation to cognitive decline. *Neuro Sci*. 2022;43(7):4107–24. <https://doi.org/10.1007/s10072-022-06068-x>.
- Hou P, Xue H, Zhang Y, Ping Y, Zheng Y, Wang Y, et al. Mediating effect of loneliness in the relationship between depressive symptoms and cognitive frailty in Community-Dwelling older adults. *Brain Sci*. 2022;12(10):1341. <https://doi.org/10.3390/brainsci12101341>.
- Nolen-Hoeksema S. Sex differences in unipolar depression: evidence and theory. *Psychol Bull*. 1987;101(2):259–82.
- Borkovec TD, Robinson E, Pruzinsky T, DeFree JA. Preliminary exploration of worry: some characteristics and processes. *Behav Res Ther*. 1983;21(1):9–16. [https://doi.org/10.1016/0005-7967\(83\)90121-3](https://doi.org/10.1016/0005-7967(83)90121-3).
- Mahoney AE, McEvoy PM, Moulds ML. Psychometric properties of the repetitive thinking questionnaire in a clinical sample. *J Anxiety Disord*. 2012;26(2):359–67. <https://doi.org/10.1016/j.janxdis.2011.12.003>.
- Ehring T, Watkins ER. Repetitive negative thinking as a transdiagnostic process. *Int J Cogn Therapy*. 2008.
- Nota JA, Coles ME. Shorter sleep duration and longer sleep onset latency are related to difficulty disengaging attention from negative emotional images in individuals with elevated transdiagnostic repetitive negative thinking. *J Behav Ther Exp Psychiatry*. 2018;58:114–22. <https://doi.org/10.1016/j.jbtep.2017.10.003>.
- Spinhoven P, van Hemert AM, Penninx BW. Repetitive negative thinking as a predictor of depression and anxiety: a longitudinal cohort study. *J Affect Disord*. 2018;241:216–25. <https://doi.org/10.1016/j.jad.2018.08.037>.
- Ehring T, Zetsche U, Weidacker K, Wahl K, Schönfeld S, Ehlers A. The perseverative thinking questionnaire (PTQ): validation of a content-independent measure of repetitive negative thinking. *J Behav Ther Exp Psychiatry*. 2011;42(2):225–32. <https://doi.org/10.1016/j.jbtep.2010.12.003>.
- Nolen-Hoeksema S, Wisco BE, Lyubomirsky S. Rethinking rumination. *Perspect Psychol Sci*. 2008;3(5):400–24. <https://doi.org/10.1111/j.1745-6924.2008.00088.x>.
- McEvoy PM, Watson H, Watkins ER, Nathan P. The relationship between worry, rumination, and comorbidity: evidence for repetitive negative thinking as a transdiagnostic construct. *J Affect Disord*. 2013;151(1):313–20. <https://doi.org/10.1016/j.jad.2013.06.014>.

23. Marchant NL, Lovland LR, Jones R, et al. Repetitive negative thinking is associated with amyloid, Tau, and cognitive decline. *Alzheimers Dement*. 2020;16(7):1054–64. <https://doi.org/10.1002/alz.12116>.
24. Schlosser M, Demnitz-King H, Whitfield T, Wirth M, Marchant NL. Repetitive negative thinking is associated with subjective cognitive decline in older adults: a cross-sectional study. *BMC Psychiatry*. 2020;20(1):500. <https://doi.org/10.1186/s12888-020-02884-7>.
25. Kelsey J, Whittemore A, Evans A, Thompson W. Methods of sampling and Estimation of sample size. *Methods Observational Epidemiol*. 1996;311:340.
26. Ai YT, Hu H, Wang L, Gao XL, Wang ZC, Ren HR et al. Current status of cognitive function and risk factors of the older adults in Wuhan. *Chinese Journal of Gerontology*. 2019;39(10):2507–2510. doi: 10. 3969/j. issn. 1005–9202. 2019. 10. 065.
27. Kornacka M, Buczny J, Layton RL. Assessing repetitive negative thinking using categorical and transdiagnostic approaches: a comparison and validation of three Polish Language adaptations of Self-Report questionnaires. *Front Psychol*. 2016;7:322. <https://doi.org/10.3389/fpsyg.2016.00322>.
28. Devynck F, Kornacka M, Baeyens C, Serra É, Neves JFD, Gaudrat B, et al. Perseverative thinking questionnaire (PTQ): French validation of a transdiagnostic measure of repetitive negative thinking. *Front Psychol*. 2017;8:2159. <https://doi.org/10.3389/fpsyg.2017.02159>.
29. China Dementia and Cognitive Disorders Guidelines Writing Group, Professional Committee of Cognitive Impairment Diseases of Chinese Medical Doctor Association Neurosurgery Branch. 2018 China dementia and cognitive disorders diagnosis and treatment guidelines (V): diagnosis and treatment of mild cognitive impairment. *Nat. Med. J. China*. 2018;98(17):1294–1301. <https://doi.org/10.3760/cma.j.issn.0376-2491.2018.17.00>
30. Zhang LX, Liu XQ. A study on reliability and validity of MOCA scale of Chinese version. *Chin Nurs Res*. 2007;31:2906–7.
31. Demnitz-King H, Göehre I, Marchant NL. The neuroanatomical correlates of repetitive negative thinking: A systematic review. *Psychiatry Res Neuroimaging*. 2021;316:111353. <https://doi.org/10.1016/j.psychres.2021.111353>.
32. Liu S, Abdellaoui A, Verweij KJH, van Wingen GA. Gene expression has distinct associations with brain structure and function in major depressive disorder. *Adv Sci (Weinh)*. 2023;10(7):e2205486. <https://doi.org/10.1002/advs.202205486>.
33. Wu JJ, Wang HX, Yao W, Yan Z, Pei JJ. Late-life depression and the risk of dementia in 14 countries: a 10-year follow-up study from the survey of health, ageing and retirement in Europe. *J Affect Disord*. 2020;274:671–7. <http://doi.org/10.1016/j.jad.2020.05.059>.
34. Lewis EJ, Blanco I, Raila H, Joormann J. Does repetitive negative thinking affect attention? Differential effects of worry and rumination on attention to emotional stimuli. *Emotion*. 2019;19(8):1450–62. <https://doi.org/10.1037/emo000535>.
35. Mennies RJ, Stewart LC, Olino TM. The relationship between executive functioning and repetitive negative thinking in youth: A systematic review of the literature. *Clin Psychol Rev*. 2021;88:102050. <https://doi.org/10.1016/j.cpr.2021.102050>.
36. Alavijeh MS, Zandiyeh Z, Moeini M. The effect of self-care self-efficacy program on life satisfaction of the Iranian elderly. *J Educ Health Promot*. 2021;10(1):167. https://doi.org/10.4103/jehp.jehp_928_20.
37. Dent E, Lien C, Lim WS, Wong WC, Wong CH, Ng TP, et al. The Asia-Pacific clinical practice guidelines for the management of frailty. *J Am Med Dir Assoc*. 2017;18(7):564–75. <https://doi.org/10.1016/j.jamda.2017.04.018>.
38. Wen W, Zhang Y, Shi W, Li J. Association between internet use and physical health, mental health, and subjective health in Middle-aged and older adults: nationally representative Cross-sectional survey in China. *J Med Internet Res*. 2023;25:e40956. <https://doi.org/10.2196/40956>.
39. Friebe J, Schmidt-Hertha B. Activities and barriers to education for elderly people. *J Contemp Educ Stud*. 2013;64:1.
40. Azizi A, Sepahvandi MA, Peyda N, Mohamadi J. Effective approach to the study of aging: grounded theory study. *Iran J Ageing*. 2016;10(4):88–101.
41. Prince M, Bryce R, Albanese E, Wimo A, Ribeiro W, Ferri CP. The global prevalence of dementia: a systematic review and metaanalysis. *Alzheimers Dement*. 2013;9:63–75. <https://doi.org/10.1016/j.jalz.2012.11.00>.
42. Zhang PD, Lv YB, Li ZH, Yin ZX, Li FR, Wang JN, et al. Age, period, and cohort effects on activities of daily living, physical performance, and cognitive functioning impairment among the Oldest-Old in China. *J Gerontol Biol Sci Med Sci*. 2020;75(6):1214–21. <https://doi.org/10.1093/gerona/glz196>.
43. Wang L, Cao M, Pu T, Huang H, Marshall C, Xiao M. Enriched physical environment attenuates spatial and social memory impairments of aged socially isolated mice. *Int J Neuropsychopharmacol*. 2018;21(12):1114–27. <https://doi.org/10.1093/ijnp/pyy084>.
44. Matyjaszek-Matuszek B, Lenart-Lipińska M, Woźniakowska E. Clinical implications of vitamin D deficiency. *Prz Menopauzalny*. 2015;14(2):75–81. <https://doi.org/10.5114/pm.2015.52149>.
45. Aihemaitijiang S, Ye C, Halimulati M, Huang X, Wang R, Zhang Z. Development and validation of nutrition literacy questionnaire for the Chinese elderly. *Nutrients*. 2022;14(5):1005. <https://doi.org/10.3390/nu14051005>.
46. Hämmig O, Bauer GF. The social gradient in work and health: a cross-sectional study exploring the relationship between working conditions and health inequalities. *BMC Public Health*. 2013;13:1170. <https://doi.org/10.1186/1471-2458-13-1170>.
47. Zhang D, Wu S, Zhang Y, Yang P, MacIntyre CR, Seale H, et al. Health literacy in Beijing: an assessment of adults' knowledge and skills regarding communicable diseases. *BMC Public Health*. 2015;15:799. <https://doi.org/10.1186/s12889-015-2151-1>.

Publisher's note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.